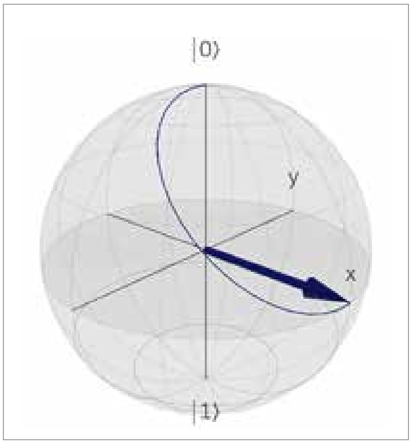

Figure 4: The effect of Hadamard gate on |0⟩

the next article itself, we will shed much more light on this idea and make the statement far less mysterious.

Installing Qiskit

Now let us begin our hands-on exploration of quantum computing by installing Qiskit, one of the most popular quantum software development kits (SDKs). Qiskit is a Python-based SDK developed by IBM for creating and running quantum circuits.

There are several ways to use Qiskit. However, one of the most operating system–agnostic and convenient approaches is to install it using Anaconda or Miniconda. Anaconda is a full Python distribution that includes many scientific computing packages. It typically requires about 2GB to 3GB of storage space. Miniconda, on the other hand, is a lightweight installer that provides only the conda package manager and Python, requiring about 400MB to 500MB of storage space.

To install Anaconda on Ubuntu, download the installer from the official website and run the following command in the terminal: *bash Anaconda3-latest-Linux-x86_64.sh*. Note that, at the time of writing this article, the latest version of Anaconda installer for Linux is *Anaconda3-2025.12-2-Linux-x86_64.sh*. To install Miniconda, download the installer from the official website and run the following command in the terminal: *bash Miniconda3-latest-Linux-x86_64.sh*. Depending on your storage space and requirements, you may choose either Anaconda or Miniconda for your system.

After installing Anaconda or Miniconda, the next step is to create a virtual environment. This step is important because the version of Python used in the environment matters when installing Qiskit. The versions of Python that work well with the latest version of Qiskit are Python 3.9, 3.10, 3.11, and 3.12. For most users, it is safest to use Python 3.10 or Python 3.11, as these versions are widely tested and stable.

Run the following commands in the terminal to create a virtual environment called *qiskit_env*, activate it, and install Qiskit. The final command will display the version of Qiskit installed on your system. In my system, the installed version of Qiskit is 2.3.0.

```
conda create -n qiskit_env python=3.11
conda activate qiskit_env
pip install qiskit
python -c "import qiskit;
print(qiskit.__version__)"
```

Our first quantum program

Now let us execute our first quantum program. The Python program titled '*qiskit1.py*' (line numbers included for clarity) creates a single qubit and applies a Hadamard gate to it. We are not measuring the value of the qubit at this stage because we will first discuss the theoretical aspects of quantum measurement in detail before performing any measurement operations.

```
1.   from qiskit import QuantumCircuit
2.   qc = QuantumCircuit(1)
3.   qc.h(0)
4.   print(qc.draw( ))
```

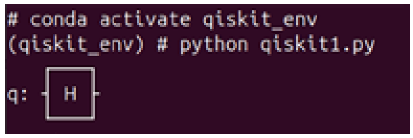

Figure 5: Output of the program *qiskit1.py*

Now let us try to understand how the program works. Line 1 imports the class QuantumCircuit from the library Qiskit. The class QuantumCircuit is used to create and manipulate quantum circuits, which are sequences of quantum operations (gates) applied to qubits. Line 2 creates a quantum circuit containing one qubit and assigns it to the variable qc. The number 1 indicates that the circuit has one qubit. When a qubit is created, it is initialised in the state |0⟩ by default. Thus, at this stage the circuit contains a single qubit in the state |0⟩. Line 3 applies a Hadamard gate to qubit 0 (the first qubit in the circuit). Here, h represents the Hadamard gate, and 0 is the index of the qubit on which the gate is applied. We have already discussed the mathematical operations and physical interpretation of the Hadamard gate. Line 4 generates and prints a visual representation of the circuit. When the program 'qiskit1.py' is executed, the output shown in Figure 5 is obtained.

In this article, we have outlined the agenda for the series and introduced some of the fundamental ideas of quantum computing, including the concept of qubits and the basic working of quantum gates. In the next article, we will explore additional quantum gates, examine their properties, and understand how measurement is performed on a qubit to extract information from a quantum system. We will also continue our journey into quantum programming with Qiskit, where we will begin building and experimenting with simple quantum circuits. END

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